

DrugUtilisation

Conduct Drug Utilisation Studies in OMOP CDM

2026-09-15

Introduction

Drug utilisation studies (DUS) were defined by the World Health Organization as studying the marketing, distribution, prescription, and use of medicinal products in a society, with special emphasis on the resulting medical and socioeconomic consequences (WHO, 2003).

This package aims to standardise and provide the tools to conduct Drug Utilisation studies as of the Darwin-EU Catalogue of Standard Analytics.



The package

```
install.packages("DrugUtilisation")
```

- **v1.2.0** on cran.
- **package website**
- **Report an issue/Feature request**

Create a reference to a cdm object

For this example we are going to use a synthetic test database:

```
library(duckdb)
library(CDMConnector)
library(dplyr)
library(CodelistGenerator)
library(DrugUtilisation)
library(CohortConstructor)
library(here)

con <- dbConnect(drv = duckdb(dbdir = here("Databases", "delphi.duckdb")))
```

```
[1] 0
```

```
cdm <- cdmFromCon(
  con = con,
  cdmSchema = "main",
  writeSchema = "results",
  writePrefix = "os_"
)
```

Add some random quantities

```
cdm$drug_exposure <- cdm$drug_exposure |>  
  mutate(quantity = as.numeric(sql("[1, 5, 10, 30, 100][floor(random() * 5)::INTEGER + 1]")))
```

Generate a drug cohort

For this example we are going to use all records of **acetaminophen**:

```
codeListPara <- getDrugIngredientCodes(cdm = cdm, name = "acetaminophen", nameStyle = "{concept_name}") |>
  subsetOnDoseUnit(cdm = cdm, doseUnit = "milligram") |>
  excludeConcepts(cdm = cdm, concepts = 1125315L, codeListName = "acetaminophen")

cdm$dus_cohort <- conceptCohort(
  cdm = cdm,
  name = "dus_cohort",
  conceptSet = codeListPara
)
```

The results is the cdm object with the new cohort instantiated.

```
cdm
```

— # OMOP CDM reference (duckdb) of Delphi-2M —

- omop tables: care_site, cdm_source, cohort_definition, concept, concept_ancestor, concept_class, concept_relationship, concept_synonym, condition_era, condition_occurrence, cost, death, device_exposure, domain, dose_era, drug_era, drug_exposure, drug_strength, fact_relationship, location, measurement, metadata, note, note_nlp, observation, observation_period, payer_plan_period, person, procedure_occurrence, provider, relationship, source_to_concept_map, specimen, visit_detail, visit_occurrence, vocabulary
- cohort tables: dus_cohort
- achilles tables: -
- other tables: -

Cohort objects

The cohort objects have some attributes:

```
class(cdm$dus_cohort)
```

```
[1] "cohort_table"      "GeneratedCohortSet" "cdm_table"         "tbl_duckdb_connection"  
[5] "tbl_dbi"           "tbl_sql"           "tbl_lazy"          "tbl"
```

```
names(attributes(cdm$dus_cohort))
```

```
[1] "names"      "class"      "tbl_source"  "tbl_name"   "cohort_set"  "cohort_attrition"  
[7] "cohort_codelist" "cdm_reference"
```

- Settings
- Counts
- Attrition
- Codelist

Cohort attributes

Settings:

```
settings(cdm$dus_cohort)
```

```
# A tibble: 1 × 4
  cohort_definition_id cohort_name    cdm_version vocabulary_version
      <int>    <chr>          <chr>          <chr>
1             1 acetaminophen 5.4          v5.0 27-AUG-25
```

Cohort counts:

```
cohortCount(cdm$dus_cohort)
```

```
# A tibble: 1 × 3
  cohort_definition_id number_records number_subjects
      <int>          <int>          <int>
1             1           338           210
```

Cohort attributes

Cohort attrition:

```
attrition(cdm$dus_cohort)
```

A tibble: 4 × 7

	cohort_definition_id	number_records	number_subjects	reason_id	reason	excluded_records	excluded_subjects
	<int>	<int>	<int>	<int>	<chr>	<int>	<int>
1	1	338	210	1	Initial qualifying e...	0	0
2	1	338	210	2	Record in observation	0	0
3	1	338	210	3	Not missing record d...	0	0
4	1	338	210	4	Merge overlapping re...	0	0

Cohort codelist:

```
cohortCodelist(cdm$dus_cohort, cohortId = 1)
```

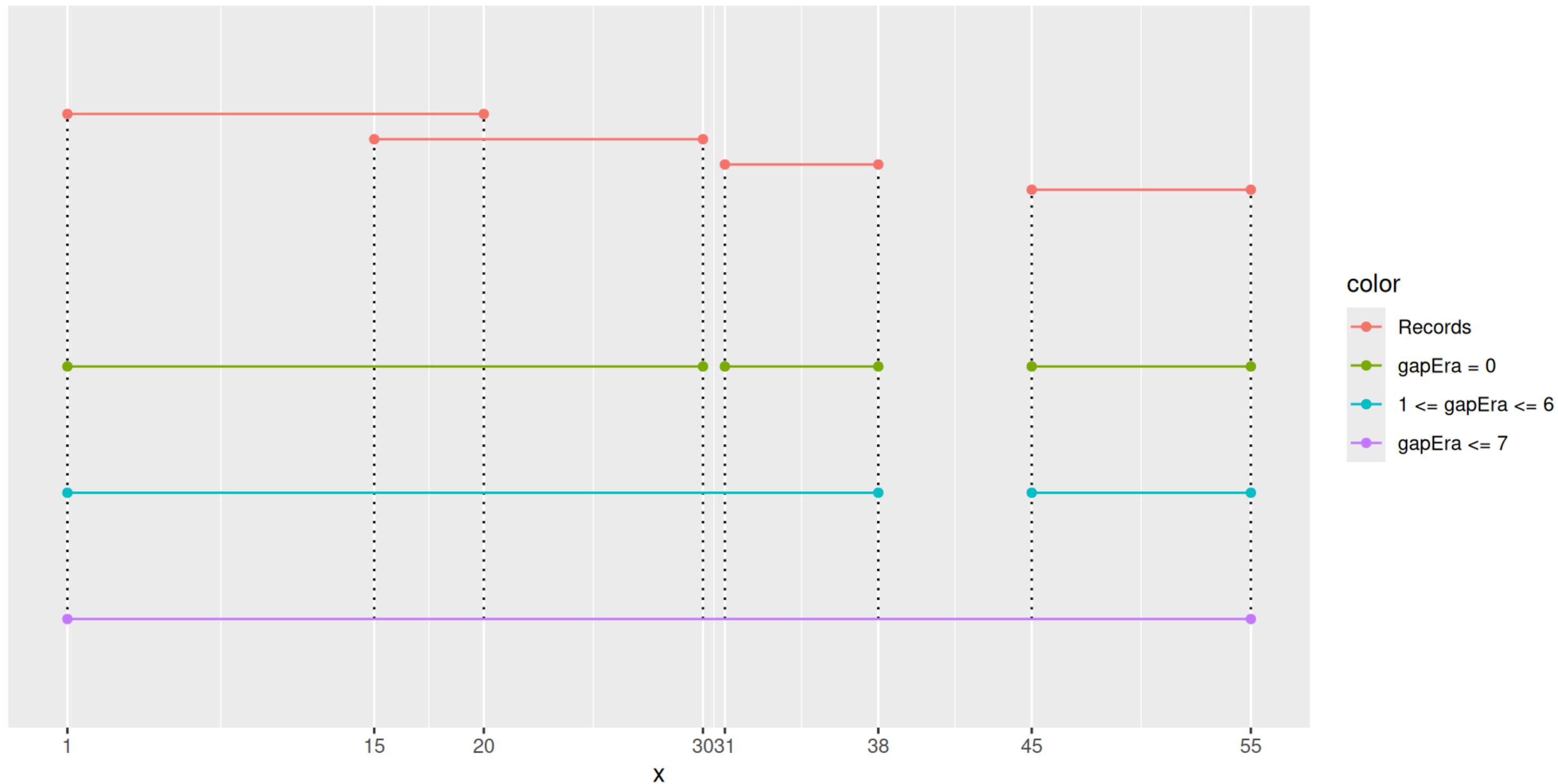
- acetaminophen (24309 codes)

gapEra parameter

gapEra parameter can be used to concatenate different records.

Number of days between two continuous exposures to be considered in the same era. Records that have fewer days between them than this gap will be concatenated into the same cohort record.

gapEra parameter



Generate a drug cohort

```
cdm$dus_cohort <- cdm$dus_cohort |>  
  collapseCohorts(gap = 30)
```

Generate a drug cohort

```
cohortCount(cdm$dus_cohort)
```

```
# A tibble: 1 × 3
```

	cohort_definition_id	number_records	number_subjects
	<int>	<int>	<int>
1	1	240	210

```
attrition(cdm$dus_cohort)
```

```
# A tibble: 5 × 7
```

	cohort_definition_id	number_records	number_subjects	reason_id	reason	excluded_records	excluded_subjects
	<int>	<int>	<int>	<int>	<chr>	<int>	<int>
1	1	338	210	1	Initial qualifying e...	0	0
2	1	338	210	2	Record in observation	0	0
3	1	338	210	3	Not missing record d...	0	0
4	1	338	210	4	Merge overlapping re...	0	0
5	1	240	210	5	Collapse cohort with...	98	0

Inclusion criteria

Four functions to add inclusion criteria to our cohort:

- Washout `requirePriorDrugWashout()` (`requireConceptIntersect`)
- First observation `requireIsFirstDrugEntry()` (`requireIsFirstEntry`)
- Prior observation `requireObservationBeforeDrug()` (`requirePriorObservation`)
- Date range `requireDrugInDateRange()` (`requireInDateRange`)

Important considerations

Not all inclusion criteria are commutable operations ($A + B \neq B + A$):

Not commutable:

- `prior washout + prior observation` $\neq$ `prior observation + prior washout`
- `prior washout + first` $\neq$ `first + prior washout`
- `prior washout + date range` $\neq$ `date range + prior washout`
- `first + prior observation` $\neq$ `prior observation + first`
- `first + date range` $\neq$ `date range + first`

Commutable:

- `date range + prior observation` $=$ `prior observation + date range`

Recommended order

Having all this into account the recommended order to apply criteria would be:

1. Require a prior drug washout or require first drug entry (particular case).
2. Require a prior observation before the drug episode.
3. Require the drugs to be in a certain date range.

Vignette to create cohorts

https://darwin-eu-dev.github.io/DrugUtilisation/articles/create_cohorts.html

Your turn

Can you create a cohort with all the records of 'ibuprofen' concatenating records separated by 90 days or less and the following inclusion criteria:

- First record ever
- At least 365 days of prior observation
- Cohort start date between '2010-01-01' and '2021-12-31'.

```
# A tibble: 1 × 3
  cohort_definition_id number_records number_subjects
      <int>           <int>           <int>
1             1             281             281
```

▶  [Click to see solution](#)

Drug usage

```
result <- cdm$dus_cohort |>
  summariseDrugUtilisation(
    ingredientConceptId = 1125315L,
    conceptSet = codelistPara,
    indexDate = "cohort_start_date",
    censorDate = "cohort_end_date",
    restrictIncident = TRUE,
    gapEra = 30,
    numberExposures = TRUE,
    numberEras = TRUE,
    daysExposed = TRUE,
    daysPrescribed = TRUE,
    timeToExposure = FALSE,
    initialQuantity = TRUE,
    cumulativeQuantity = TRUE,
    initialDailyDose = TRUE,
    cumulativeDose = TRUE,
    estimates = c("q25", "median", "q75")
```

Drug usage

```
result |>
  suppress(minCellCount = 5) |>
  tableDrugUtilisation()
```

Concept set	Ingredient	Variable name	Estimate name	Data source
				Delphi-2M
acetaminophen				
overall	overall	number records	N	240
		number subjects	N	210
acetaminophen	overall	number exposures	Median (Q25 - Q75)	1 (1 - 2)
		cumulative quantity	Median (Q25 - Q75)	15.00 (5.00 - 40.00)
		initial quantity	Median (Q25 - Q75)	10.00 (5.00 - 30.00)
		initial exposure duration	Median (Q25 - Q75)	1 (1 - 1)
		number eras	Median (Q25 - Q75)	1 (1 - 1)
		days exposed	Median (Q25 - Q75)	1 (1 - 23)
		days prescribed	Median (Q25 - Q75)	1 (1 - 21)
	acetaminophen	cumulative dose milligram	Median (Q25 - Q75)	5,500.00 (2,500.00 - 20,000.00)
		initial daily dose milligram	Median (Q25 - Q75)	5,000.00 (2,500.00 - 15,000.00)

Indications

`summariseIndication()`:

- To summarise **mutually exclusive** indications.
- Define a window respect to the 'cohort_start_date'.
- Indications must be instantiated beforehand as cohorts.
- Unknown indication (check a table to see if there is a record).

Indications

Let's instantiate the cohorts of interest:

```
cdm$indications <- conceptCohort(  
  cdm = cdm,  
  conceptSet = list(  
    osteoarthritis = 4014829L,  
    fever = 4019097L,  
    pain = c(35953849L, 35947730L, 40231906L)  
  ),  
  name = "indications"  
)
```

Indciations

```

result <- cdm$dus_cohort |>
  summariseIndication(
    indicationCohortName = "indications",
    indicationWindow = list(c(0, 0), c(-30, 7)),
    unknownIndicationTable = "condition_occurrence"
  )
result |>
  glimpse()

```

Rows: 36

Columns: 13

```

$ result_id      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ...
$ cdm_name       <chr> "Delphi-2M", "Delphi-2M", "Delphi-2M", "Delphi-2M", "Delphi-2M", "Delphi-2M", "Delphi-2M", "D...
$ group_name     <chr> "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "co...
$ group_level    <chr> "acetaminophen", "acetaminophen", "acetaminophen", "acetaminophen", "acetaminophen", "acetaminophen", "acetami...
$ strata_name    <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over...
$ strata_level   <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over...
$ variable_name  <chr> "Indication on index date", "Indication on index date", "Indication on index date", "Indicati...
$ variable_level <chr> "fever", "fever", "osteoarthritis", "osteoarthritis", "pain", "pain", "fever and osteoarthritis...
$ estimate_name  <chr> "count", "percentage", "count", "percentage", "count", "percentage", "count", "percentage", "...
$ estimate_type  <chr> "integer", "percentage", "integer", "percentage", "integer", "percentage", "integer", "percen...
$ estimate_value <chr> "0", "0", "77", "32.08333", "7", "2.91667", "0", "0", "19", "7.91667", "0", "0", "6", "2.5000...
$ additional_name <chr> "window_name", "window_name", "window_name", "window_name", "window_name", "window_name", "window_name", "wi...

```

Indications

```
result |>
  suppress() |>
  tableIndication()
```

			Data source
			Delphi-2M
Restrict incident	Indication	Estimate name	Cohort name
			acetaminophen
Indication on index date			
FALSE	fever	N (%)	0 (0.00 %)
	osteoarthritis	N (%)	77 (32.08 %)
	pain	N (%)	7 (2.92 %)
	fever and osteoarthritis	N (%)	0 (0.00 %)
	fever and pain	N (%)	19 (7.92 %)
	osteoarthritis and pain	N (%)	0 (0.00 %)
	fever and osteoarthritis and pain	N (%)	6 (2.50 %)
	unknown	N (%)	0 (0.00 %)
	none	N (%)	131 (54.58 %)
Indication from 30 days before to 7 days after the index date			
FALSE	fever	N (%)	0 (0.00 %)
	osteoarthritis	N (%)	101 (42.08 %)
	pain	N (%)	7 (2.92 %)
	fever and osteoarthritis	N (%)	0 (0.00 %)
	fever and pain	N (%)	19 (7.92 %)
	osteoarthritis and pain	N (%)	0 (0.00 %)

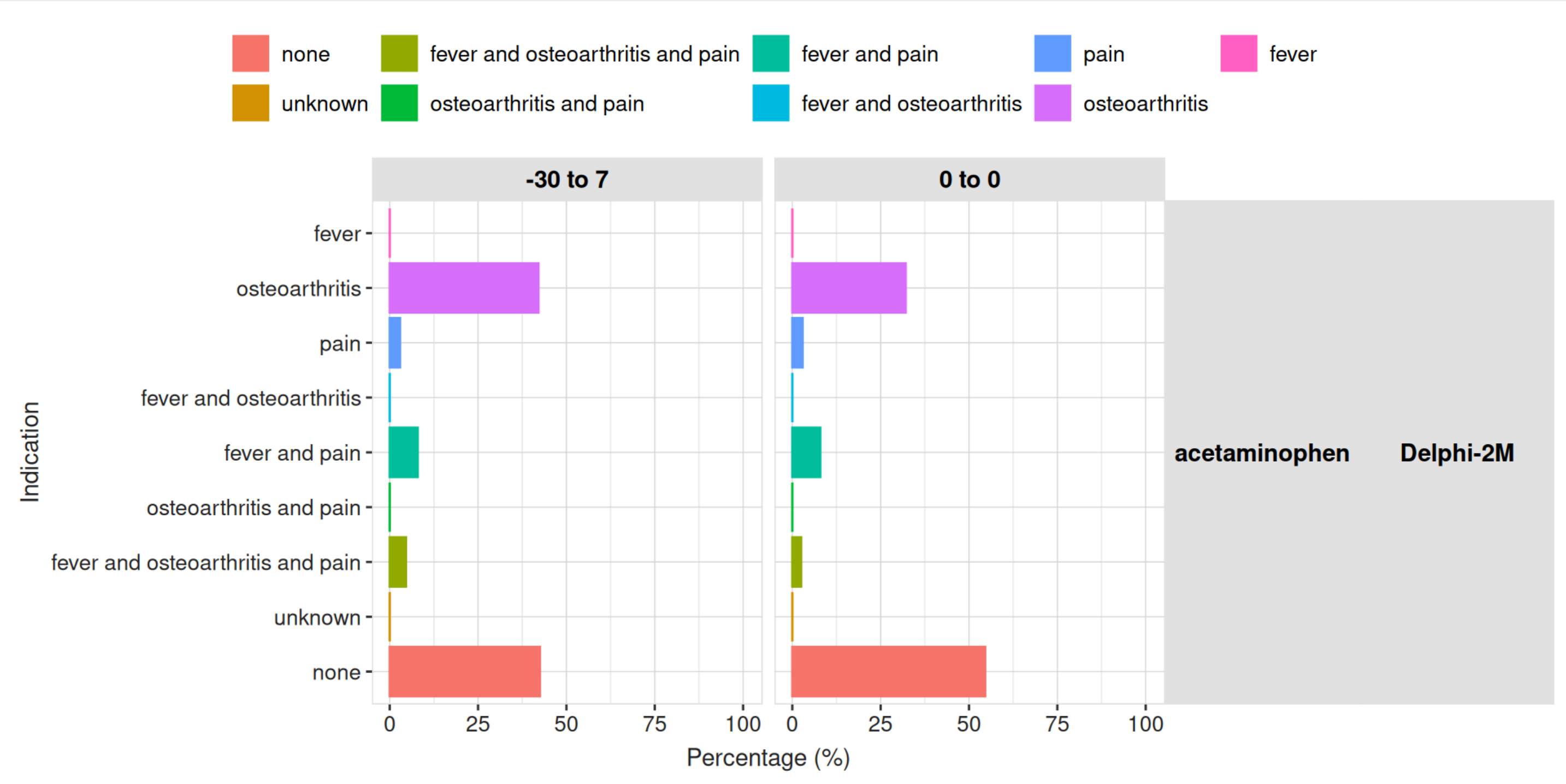
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			Data source
			Delphi-2M
Restrict incident	Indication	Estimate name	Cohort name
			acetaminophen
	fever and osteoarthritis and pain	N (%)	11 (4.58 %)
	unknown	N (%)	0 (0.00 %)
	none	N (%)	102 (42.50 %)

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Indications

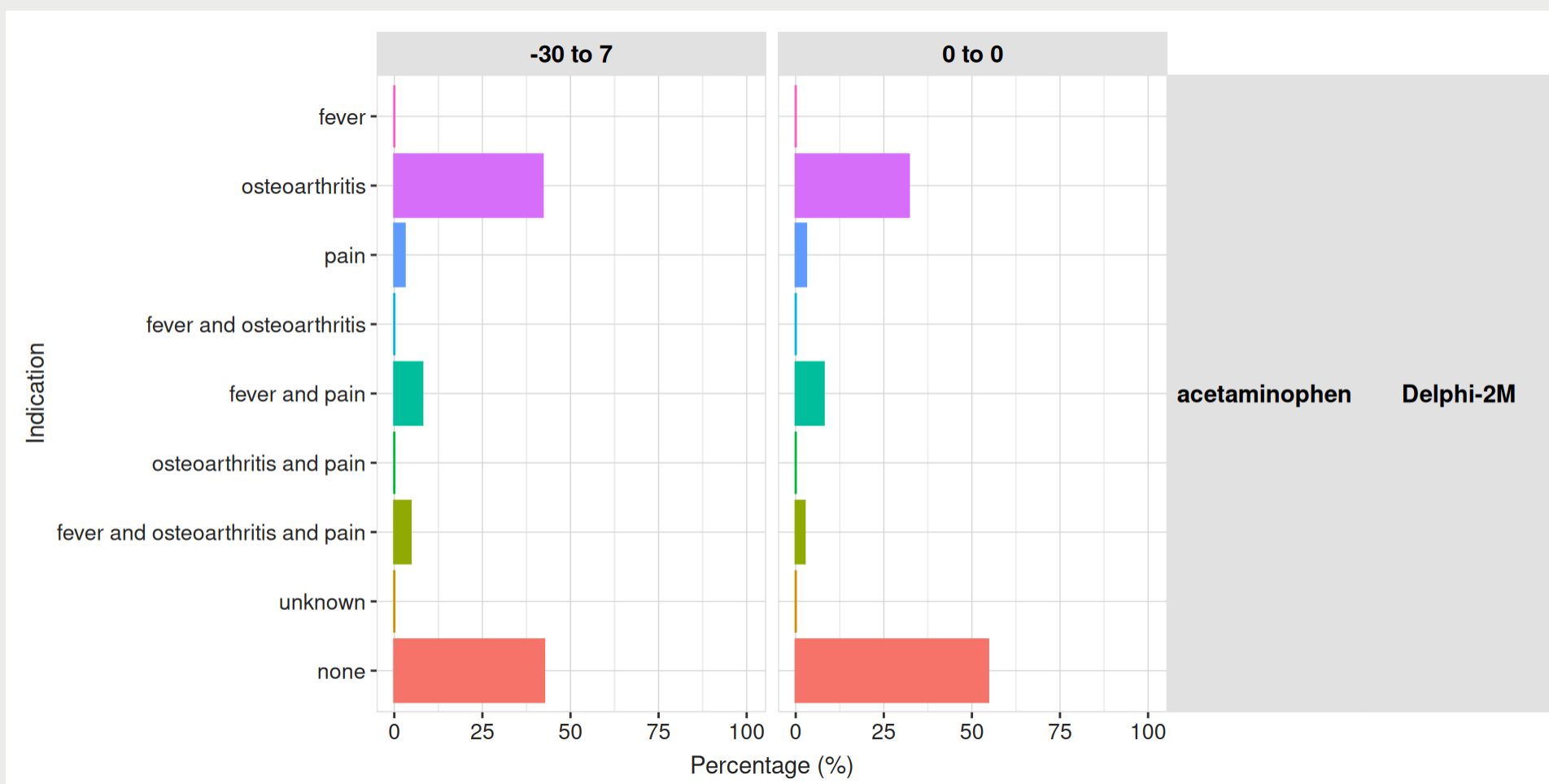
```
plotIndication(result)
```



Indications

Output of plotting functions are **ggplot2** objects.

```
plotIndication(result) +  
  ggplot2::theme(legend.position = "none")
```



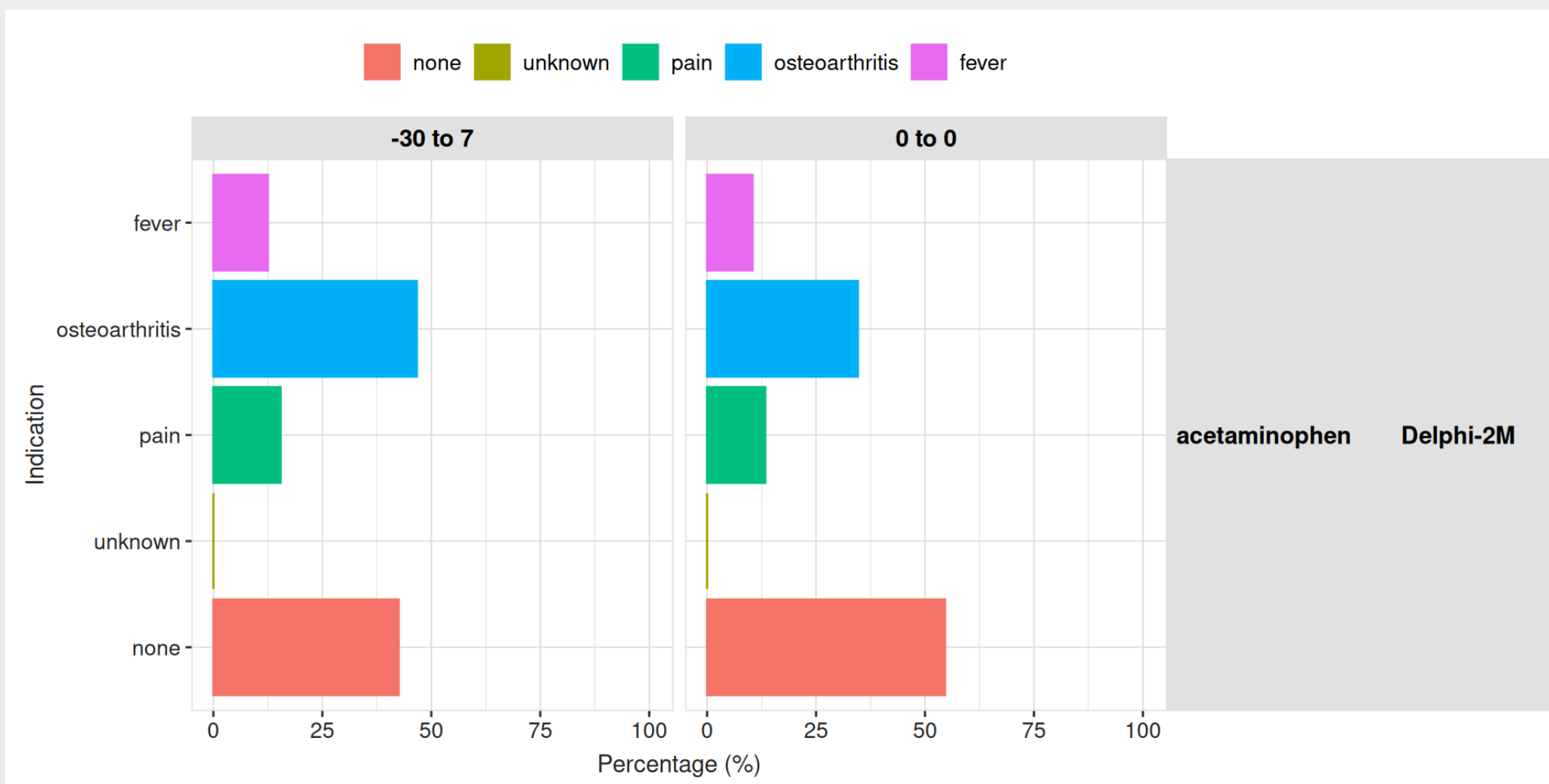
Indications

Currently considering to add support for **plotly**.

```
plotIndication(result) |>  
  plotly::ggplotly()
```

Indciations (Mutually exclusive FALSE)

```
result <- cdm$dus_cohort |>
  summariseIndication(
    indicationCohortName = "indications",
    indicationWindow = list(c(0, 0), c(-30, 7)),
    mutuallyExclusive = FALSE,
    unknownIndicationTable = "condition_occurrence"
  )
plotIndication(result = result)
```



Treatment persistence

To analyse treatment persistence and adherence there are two standard pipelines:

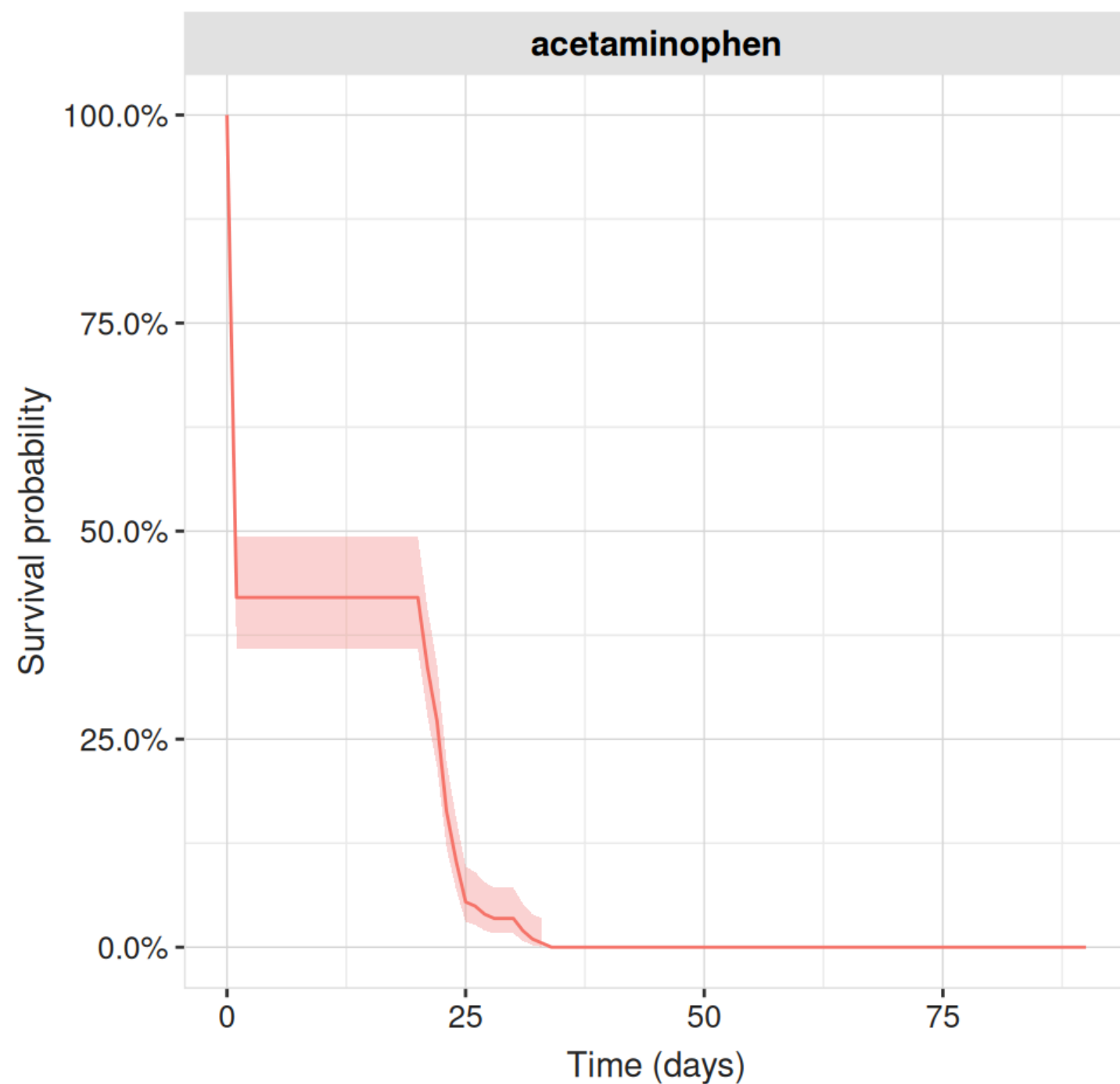
- Survival analysis
- Proportion of patients covered

Survival discontinuation

```
result <- cdm$dus_cohort |>  
  summariseDiscontinuationAsSurvival(followUpDays = 90)
```

Survival discontinuation

```
plotDiscontinuationAsSurvival(result = result)
```



variable_name

— Survival probability of discontinuation_of_acetaminophen (Outcome)

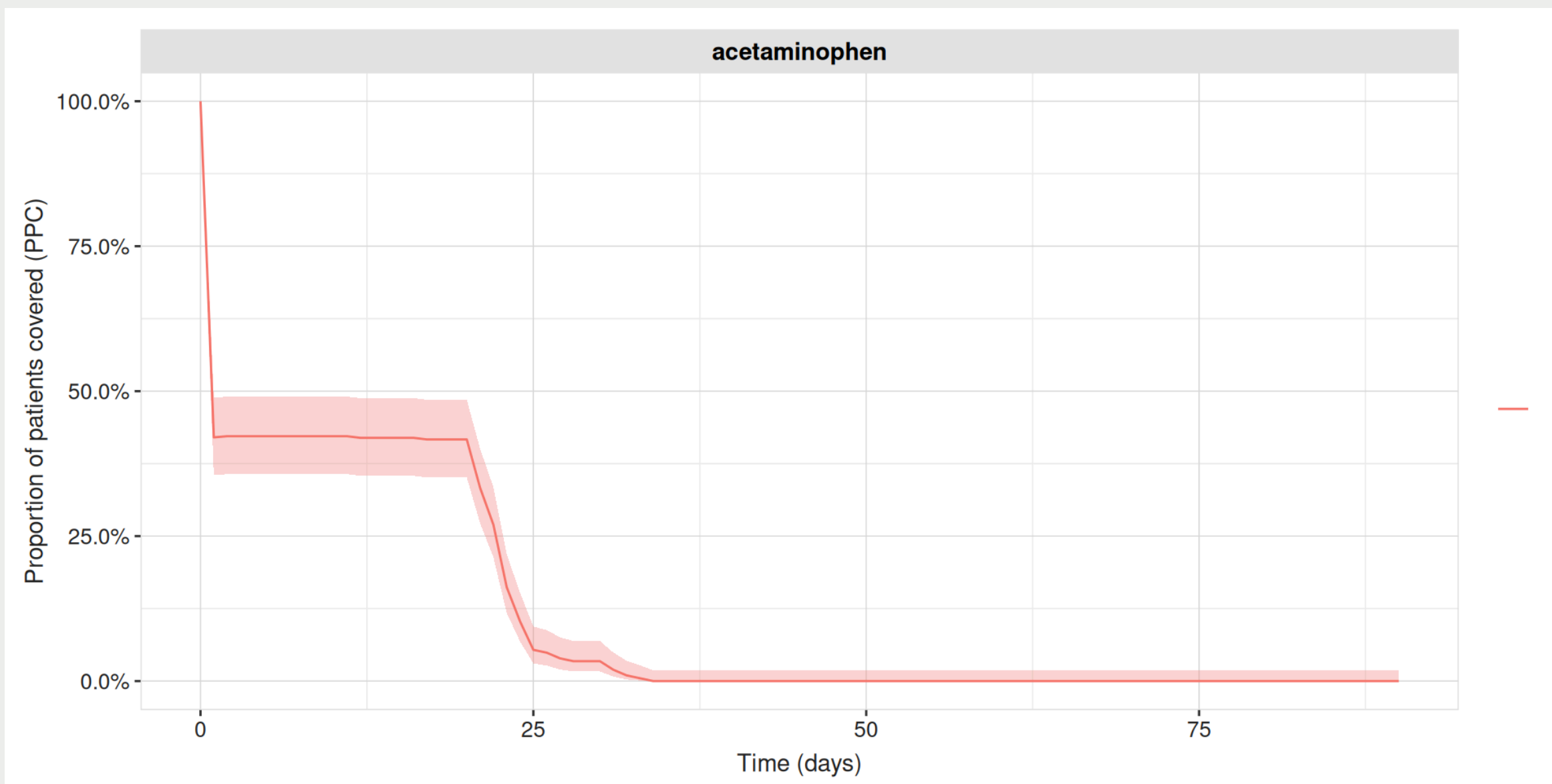
Proportion of patients covered

Proportion of treated individuals between index date and **followUpDays**. Percentages will be reported daily.

```
result <- cdm$dus_cohort |>  
  summariseProportionOfPatientsCovered(followUpDays = 90)
```

Proportion of patients covered

```
plotProportionOfPatientsCovered(result)
```

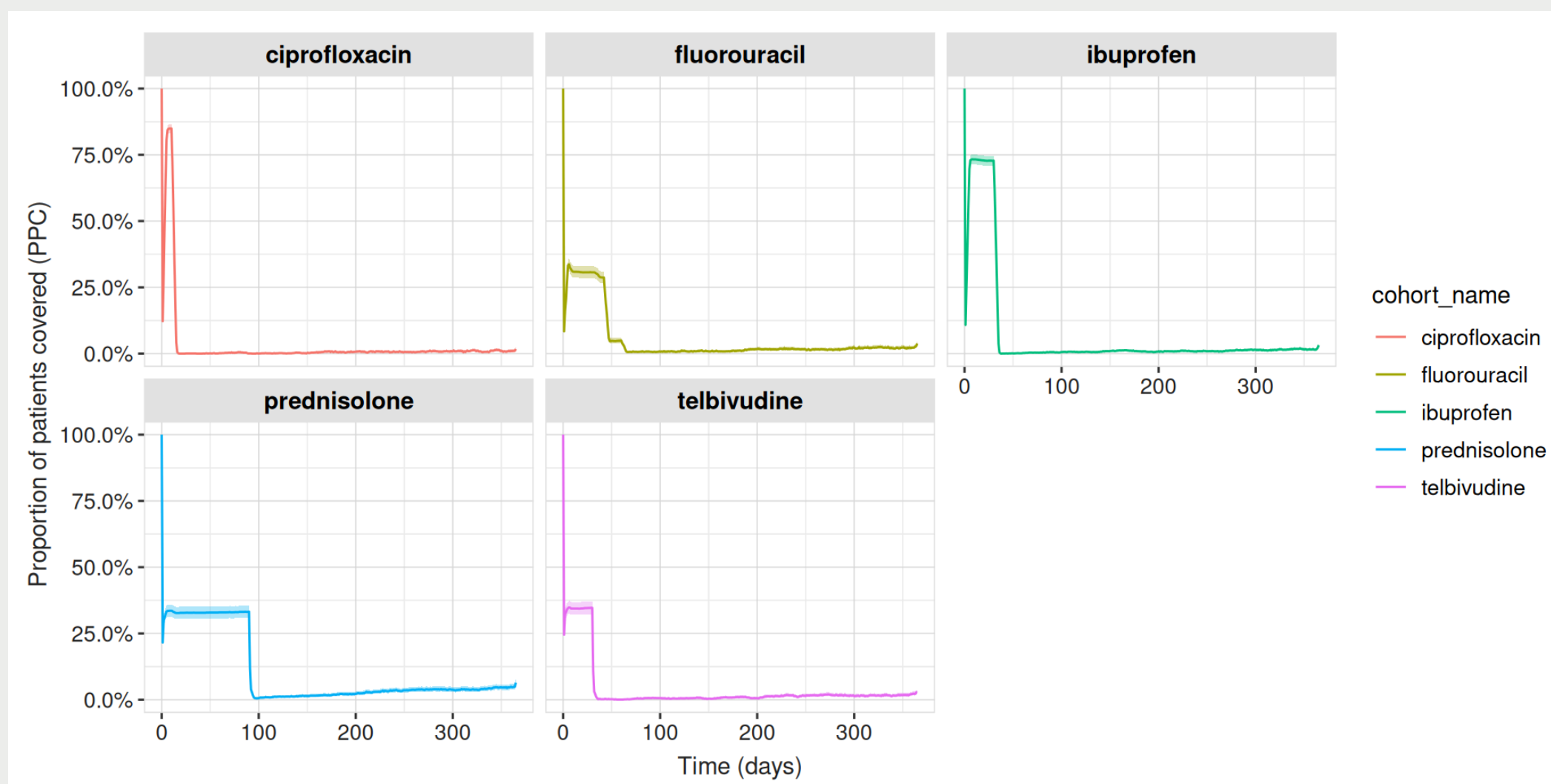


gapEra parameter can have a big impact in survival analysis, whereas it does not have such big impact on proportion of patients covered.

Your turn

Can you create a cohort of a certain ingredient and analyse its persistence in the following year?

suggestions: 'fluorouracil', 'prednisolone', 'telbivudine', 'ciprofloxacin', 'ibuprofen' are common ingredients in this database



Solution

▶  Click to see solution

Drug restart and drug switching

We have the ability to study drug restart or drug switching after a discontinuation:

- Switching cohorts must be defined in advance.
- We have to define the windows of interest.
- Index date will be the date of discontinuation.
- Reported percentages in each window will be:
- *Restart*: individuals that restarted and not switched.
- *Switch*: individuals that switched and not restarted.
- *Restart and switch*: individuals that restarted and switched.
- *Untreated*: individuals that did not restart or switch.

Drug restart and drug switching

Let's define the cohorts of interest for switching:

```
codelist <- getDrugIngredientCodes(  
  cdm = cdm,  
  name = c('fluorouracil', 'prednisolone', 'telbivudine', 'ciprofloxacin', 'ibuprofen'),  
  nameStyle = "{concept_name}"  
)  
cdm$switch <- conceptCohort(  
  cdm = cdm,  
  name = "switch",  
  conceptSet = codelist  
)
```

Drug restart and drug switching

We have to be careful with the inclusion criteria of the input cohort, as restart is derived from it.

```
result <- cdm$dus_cohort |>
  summariseDrugRestart(
    switchCohortTable = "switch",
    followUpDays = c(90, 180, 270, 360),
    restrictToFirstDiscontinuation = TRUE
  )
result |>
  glimpse()
```

Rows: 32

Columns: 13

```
$ result_id      <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1
$ cdm_name       <chr> "Delphi-2M", "Delphi-2M", "Delphi-2M", "Delphi-2M", "Delphi-2M", "Delphi-2M", "Delphi-2M", "D...
$ group_name     <chr> "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "coho...
$ group_level    <chr> "acetaminophen", "acetaminophen", "acetaminophen", "acetaminophen", "acetaminophen", "acetami...
$ strata_name    <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over...
$ strata_level   <chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over...
$ variable_name  <chr> "Drug restart in 90 days", "Drug restart in 90 days", "Drug restart in 90 days", "Drug restar...
$ variable_level <chr> "restart", "restart", "switch", "switch", "restart and switch", "restart and switch", "untrea...
$ estimate_name  <chr> "count", "percentage", "count", "percentage", "count", "percentage", "count", "percentage", "...
$ estimate_type  <chr> "integer", "percentage", "integer", "percentage", "integer", "percentage", "integer", "percen...
$ estimate_value <chr> "0", "0", "5", "2.38095238095238", "0", "0", "205", "97.6190476190476", "0", "0", "8", "3.809...
$ additional_name <chr> "follow_up_days", "follow_up_days", "follow_up_days", "follow_up_days", "follow_up_days", "fo..."
```

Drug restart and drug switching

tableDrugRestart(result)

		Data source
		Delphi-2M
Treatment	Estimate name	Cohort name
		acetaminophen
Drug restart in 90 days		
restart	N (%)	0 (0.00 %)
switch	N (%)	5 (2.38 %)
restart and switch	N (%)	0 (0.00 %)
untreated	N (%)	205 (97.62 %)
Drug restart in 180 days		
restart	N (%)	0 (0.00 %)
switch	N (%)	8 (3.81 %)
restart and switch	N (%)	0 (0.00 %)
untreated	N (%)	202 (96.19 %)
Drug restart in 270 days		
restart	N (%)	0 (0.00 %)
switch	N (%)	13 (6.19 %)
restart and switch	N (%)	0 (0.00 %)
untreated	N (%)	197 (93.81 %)
Drug restart in 360 days		
restart	N (%)	2 (0.95 %)
switch	N (%)	22 (10.48 %)
restart and switch	N (%)	0 (0.00 %)

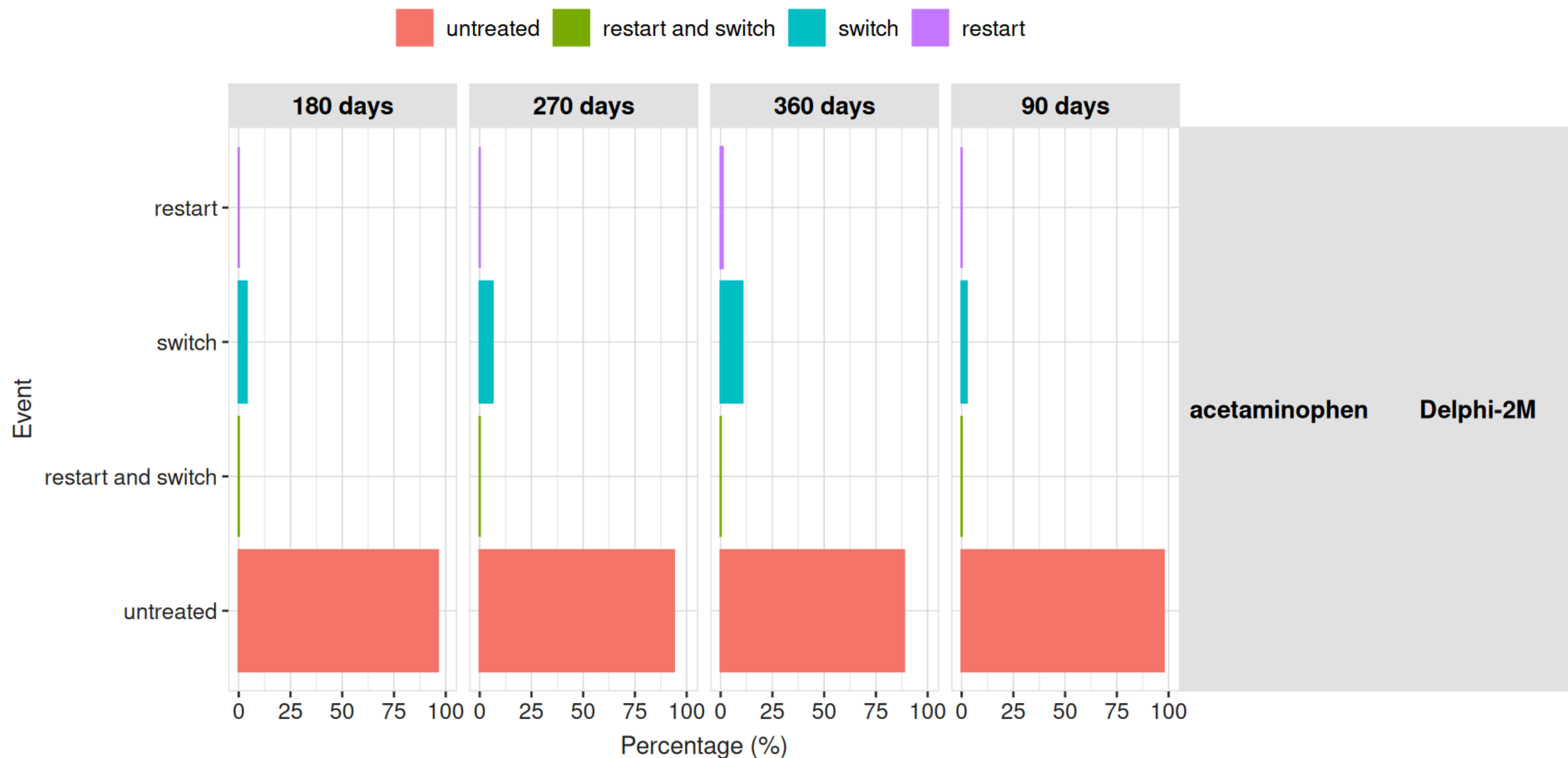
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		Data source
		Delphi-2M
Treatment	Estimate name	Cohort name
		acetaminophen
untreated	N (%)	186 (88.57 %)

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Drug restart and drug switching

```
plotDrugRestart(result)
```



Summarise treatments

`summariseTreatment()` is a general function to analyse presence of treatments (cohorts) after an index date. This can be used with many different purposes:

- Summarise treatments after a certain condition
- Analyse comedications
- Analyse treatments after discontinuation
- ...

Summarise treatments

You have to instantiate the treatments that you are interested in a cohort:

```
codelist <- getDrugIngredientCodes(  
  cdm = cdm,  
  name = c('fluorouracil', 'prednisolone', 'telbivudine', 'ciprofloxacin', 'ibuprofen'),  
  nameStyle = "{concept_name}"  
)  
cdm$treatments <- conceptCohort(  
  cdm = cdm,  
  name = "treatments",  
  conceptSet = codelist  
)  
result <- cdm$dus_cohort |>  
  summariseTreatment(  
    window = list(c(1, 90), c(91, 180), c(181, 270), c(271, 360)),  
    treatmentCohortName = "treatments",  
    indexDate = "cohort_end_date",  
    censorDate = NULL  
  )  
result |>
```

Rows: 48

Columns: 13

\$ result_id	<int> 1, ...
\$ cdm_name	<chr> "Delphi-2M", "Delphi-2M", "Delphi-2M", "Delphi-2M", "Delphi-2M", "Delphi-2M", "Delphi-2M", "D...
\$ group_name	<chr> "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "cohort_name", "co...
\$ group_level	<chr> "acetaminophen", "acetaminophen", "acetaminophen", "acetaminophen", "acetaminophen", "acetaminophen", "acetami...
\$ strata_name	<chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over...
\$ strata_level	<chr> "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "overall", "over...
\$ variable_name	<chr> "Medication from 1 day after to 90 days after the index date", "Medication from 1 day after t...

Summarise treatments

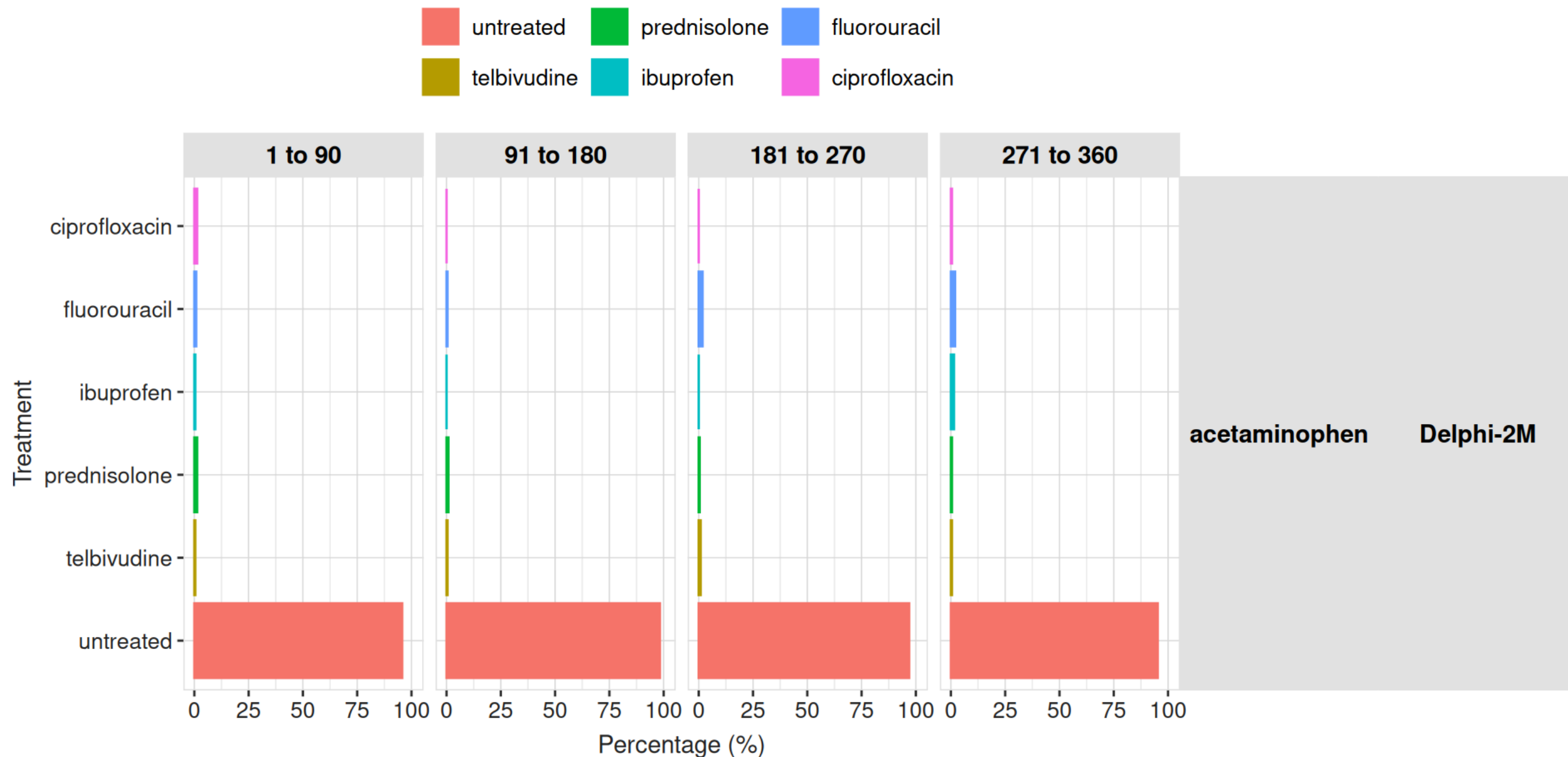
```
tableTreatment(result)
```

			Data source
			Delphi-2M
			Cohort name
Restrict incident	Treatment	Estimate name	acetaminophen
Medication from 1 day after to 90 days after the index date			
FALSE	ciprofloxacin	N (%)	3 (1.28 %)
	fluorouracil	N (%)	2 (0.85 %)
	ibuprofen	N (%)	1 (0.43 %)
	prednisolone	N (%)	3 (1.28 %)
	telbivudine	N (%)	1 (0.43 %)
	untreated	N (%)	224 (95.73 %)
Medication from 91 days after to 180 days after the index date			
FALSE	ciprofloxacin	N (%)	0 (0.00 %)
	fluorouracil	N (%)	1 (0.43 %)
	ibuprofen	N (%)	0 (0.00 %)
	prednisolone	N (%)	2 (0.87 %)
	telbivudine	N (%)	1 (0.43 %)
	untreated	N (%)	227 (98.27 %)
Medication from 181 days after to 270 days after the index date			
FALSE	ciprofloxacin	N (%)	0 (0.00 %)
	fluorouracil	N (%)	4 (1.76 %)
	ibuprofen	N (%)	0 (0.00 %)
	prednisolone	N (%)	1 (0.44 %)

			Data source
			Delphi-2M
Restrict incident	Treatment	Estimate name	Cohort name
			acetaminophen
	telbivudine	N (%)	2 (0.88 %)
	untreated	N (%)	220 (96.92 %)
Medication from 271 days after to 360 days after the index date			
FALSE	ciprofloxacin	N (%)	1 (0.47 %)
	fluorouracil	N (%)	4 (1.90 %)
	ibuprofen	N (%)	3 (1.42 %)
	prednisolone	N (%)	1 (0.47 %)
	telbivudine	N (%)	1 (0.47 %)
	untreated	N (%)	201 (95.26 %)

Summarise treatments

```
plotTreatment(result)
```



DrugUtilisation

👉 Packages website

👉 CRAN link

👉 Manual

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